

Class 10 Physics Light Reflection and Refraction

Class 10 Physics Light Reflection and Refraction explain how light behaves when it bounces off surfaces or passes from one medium to another.

It covers laws of reflection and refraction, image formation by mirrors and lenses, and practical applications in everyday life.

Reflection of Light

Reflection occurs when light hits a polished surface, like a mirror, and bounces back into the same medium.

The Laws of Reflection are:

- The angle of incidence equals the angle of reflection (angle i = angle r).
- The incident ray, the reflected ray, and the normal to the surface at the point of incidence all lie in the same plane.

Spherical Mirrors

Spherical mirrors are curved mirrors that form part of a sphere. They are of two types: concave and convex. Key terms associated with them include:

- Pole (P): The centre point of the mirror's reflecting surface.
- Centre of Curvature (C): The centre of the sphere from which the mirror is a part.
- Principal Axis (PA): The straight line passing through the Pole and the Centre of Curvature.
- Radius of Curvature (R): The distance between the Pole and the Centre of Curvature ($R = 2f$).
- Principal Focus (F): The point on the principal axis where parallel rays converge (concave) or appear to diverge from (convex) after reflection.
- Focal Length (f): The distance between the Pole and the Principal Focus.

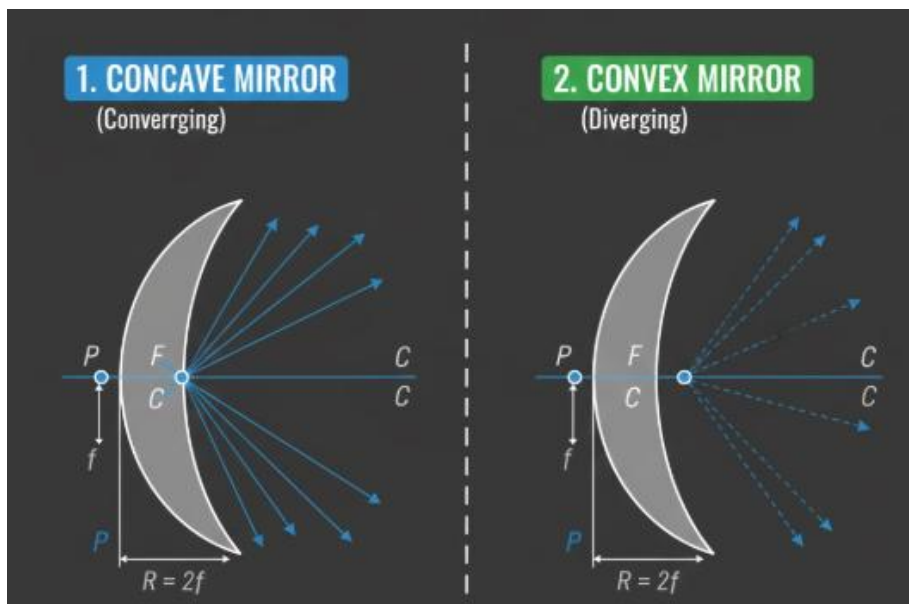
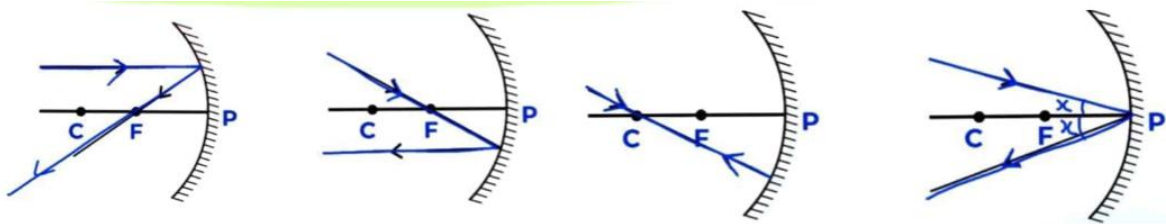


Image Formation by Concave Mirror

Concave mirrors can form both real and virtual images depending on the object's position. Real images are formed when reflected rays actually meet and are typically inverted. Virtual images form when rays appear to meet and are always erect.



Position of the Object	Position of the Image	Size of the Image	Nature of the Image
At infinity	At the focus F	Highly diminished, point-sized	Real and inverted
Beyond C	Between F and C	Diminished	Real and inverted
At C	At C	Same size	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At F	At infinity	Highly enlarged	Real and inverted
Between P and F	Behind the mirror	Enlarged	Virtual and erect

Uses of Concave Mirror

Concave mirrors are used in various applications due to their converging property.

- Shaving mirrors to see magnified images.
- Dentist's mirrors to examine teeth.
- Headlights of cars and searchlights to produce powerful parallel beams.
- Solar furnaces to concentrate sunlight.

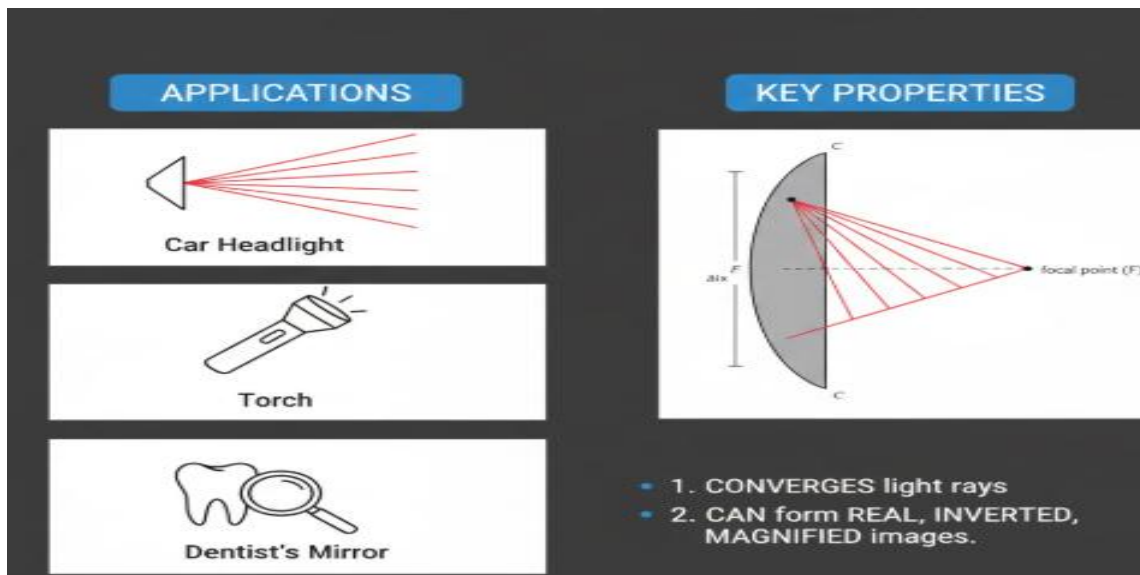
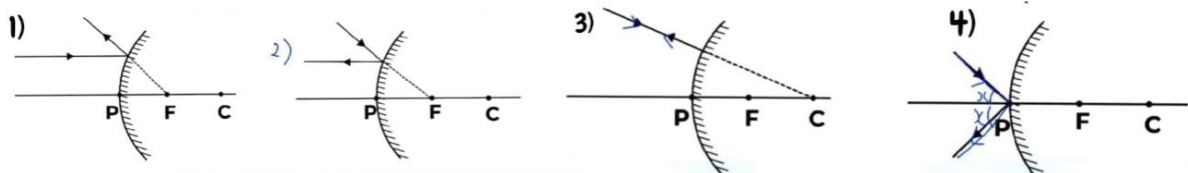


Image Formation by Convex Mirror

Convex mirrors always form virtual, erect, and diminished images, regardless of the object's position.

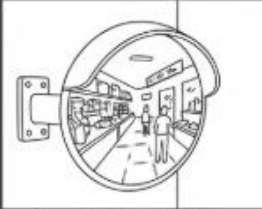


Position of the Object	Position of the Image	Size of the Image	Nature of the Image
At infinity	At the focus F	Highly diminished, point-sized	Virtual and erect
Between infinity and the pole P	Between F and P	Diminished	Virtual and erect

Uses of Convex Mirror

Convex mirrors provide a wider field of view.

- Rear-view mirrors in vehicles.
- Security mirrors in shops.

KEY PROPERTY	APPLICATIONS	SUMMARY
 1. DIVERGES light rays 2. PROVIDES a WIDER WIDER FIELD OF VIEW	 Car Side-View Mirror  Image is Small & Erect  Security Mirror in Shops	1. Provides a wider field of view. 2. Always forms a virtual, erect, and diminished image

Sign Convention for Spherical Mirrors

The Cartesian Sign Convention is used for mirror calculations.

- All distances are measured from the Pole (P).
- Distances measured in the direction of incident light are positive; those against are negative.
- Heights measured upward (above principal axis) are positive; downward are negative.

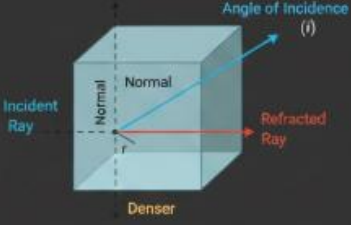
Refraction of Light

Refraction is the phenomenon of light bending as it travels from one transparent medium to another.

This bending occurs because the speed of light changes when it moves from one medium to another.

REFRACTION OF LIGHT

1. LIGHT BENDS



Light bends when passing from one medium of another.

2. THE MECHANISM

1. Caused by a change in **speed**.
3. Light travels slower in denser materials.

- The amount of bending is described by Snell's Law:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

n : Refractive Index θ : Angle



Refractive Index

The refractive index (n) of a medium quantifies how much light bends when entering it.

- Absolute Refractive Index:
where c is the speed of light in vacuum (3×10^8 m/s) and V is the speed of light in the medium.
- Relative Refractive Index: The refractive index of medium 2 with respect to medium 1 is a higher refractive index means a denser medium and slower light speed.

Laws of Refraction (Snell's Law)

The Laws of Refraction describe how light behaves when passing between two media.

- The incident ray, the refracted ray, and the normal to the interface at the point of incidence all lie in the same plane.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for a given pair of media. This is Snell's Law:

Spherical Lenses

Lenses are transparent materials bounded by two spherical surfaces or one spherical and one plane surface.

- Convex Lens (Converging Lens): Thicker in the middle, converges parallel light rays.

- Concave Lens (Diverging Lens): Thinner in the middle, diverges parallel light rays.
- Optical Centre (O): The central point of the lens through which light passes undeviated.
- Principal Focus (F) and Focal Length (f): Similar to mirrors, but lenses have two focal points (F₁, F₂) on either side.

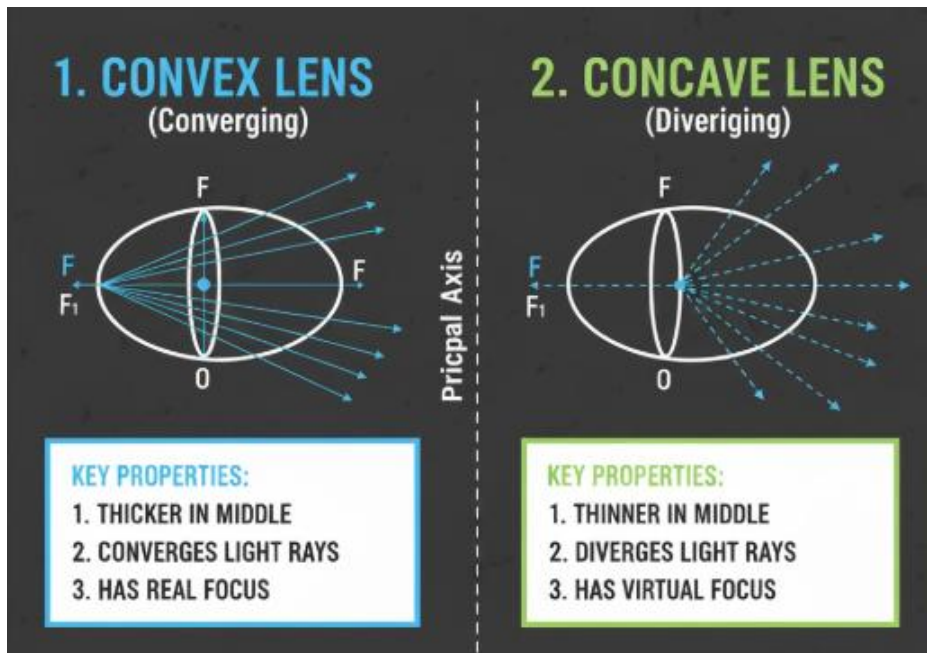


Image Formation by Convex Lens

Convex lenses form images with varied characteristics, depending on the object's position. This topic is key in class 10 physics light reflection and refraction.

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F ₂	Highly diminished, point-sized	Real and inverted
Beyond 2F ₁	Between F ₂ and 2F ₂	Diminished	Real and inverted
At 2F ₁	At 2F ₂	Same size	Real and inverted
Between F ₁ and 2F ₁	Beyond 2F ₂	Enlarged	Real and inverted
At focus F ₁	At infinity	Infinitely large or highly enlarged	Real and inverted
Between F ₁ and optical centre O	On the same side of the lens as the object	Enlarged	Virtual and erect

Image Formation by Concave Lens

Concave lenses always form virtual, erect, and diminished images.

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
Between infinity and optical centre O	Between focus F_1 and optical centre O	Diminished	Virtual and erect

Sign Convention for Spherical Lenses

The Cartesian Sign Convention is adapted for lenses:

- All distances are measured from the Optical Centre (O).
- Directions and heights follow the same rules as for mirrors.

Key Rules of Light: Reflection and Refraction

These formulas are vital for solving numerical problems in light reflection and refraction class 10.

Mirror Formula

The relationship between object distance (u), image distance (v), and focal length (f) for spherical mirrors is:

- Note: Object distance (u) is always taken as negative. Convex mirror has positive ' f ', concave mirror has negative ' f '.

Magnification (Mirrors)

Magnification (m) describes the relative size of the image to the object.

- h_i : height of image, h_o : height of object.
- Positive m indicates virtual and erect image.
- Negative m indicates real and inverted image.
- $|m| > 1$ means enlarged, $|m| < 1$ means diminished, $|m| = 1$ means same size.

Lens Formula

The relationship between object distance (u), image distance (v), and focal length (f) for spherical lenses is:

- Note: Convex lens has positive ' f ', concave lens has negative ' f '.

Magnification (Lenses)

For lenses, magnification (m) is given by:

- Positive m indicates virtual and erect image.
- Negative m indicates real and inverted image.

Power of a Lens

The power of a lens (P) measures its ability to converge or diverge light rays.

(where f is in meters).

- The unit of power is Dioptre (D).
- A convex lens has positive power; a concave lens has negative power.
- For a combination of lenses,